

Basis-mixture extremals for the sine transform

A partial answer to Maresch–Schuster (arXiv:1207.7266)

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Status. Partial result, likely valid, subject to human review.

1 Source conjecture

For a finite Borel measure μ on S^{n-1} , its sine transform is

$$(\mathcal{S}\mu)(x) = \int_{S^{n-1}} \sqrt{\|x\|^2 - (x \cdot u)^2} d\mu(u).$$

If μ is even and isotropic, $\int u \otimes u d\mu(u) = I_n$, this is the support function of a centrally symmetric convex body denoted S_μ ; its polar is S_μ^* . For an orthonormal basis $E = (e_1, \dots, e_n)$, write

$$\nu_E = \frac{1}{2} \sum_{i=1}^n (\delta_{e_i} + \delta_{-e_i})$$

for the associated cross measure. On page 14, after Theorem 4.3, Maresch and Schuster conjecture:

Among even isotropic measures, $V(S_\mu^*)$ is maximized precisely by cross measures, while $V(S_\mu)$ is minimized precisely by cross measures.

In view of Theorem [4.3](#) we formulate the following

Conjecture. *Among even isotropic measures, $V(S_\mu^*)$ is maximized precisely by cross measures, while $V(S_\mu)$ is minimized precisely by cross measures.*

2 New partial theorem

Call an even measure an *orthogonal-basis mixture* if there is a Borel probability measure π on $O(n)$ such that

$$\mu = \int_{O(n)} R_{\#} \nu_E d\pi(R). \tag{1}$$

This is exactly the closed convex hull of the cross measures. It includes finite and continuous mixtures (for example normalized spherical Lebesgue measure), but is a proper subclass of the even isotropic measures.

Theorem 1 (Both conjectured inequalities on the basis-mixture class). *Let $n \geq 3$, let ν_E be a cross measure, and let μ have the representation (1). Then*

$$V(S_\mu^*) \leq V(S_{\nu_E}^*) \quad \text{and} \quad V(S_\mu) \geq V(S_{\nu_E}).$$

Equality in either inequality holds if and only if μ itself is a cross measure.

3 Proof intuition

All cross measures are rotations of one model, so all their sine-transform bodies are rotated copies of one body K_0 . Because the sine transform is linear in the measure, S_μ is the Minkowski average of these rotated copies. Volume to the power $1/n$ grows under Minkowski averaging, which gives the minimum of the primal volume. Polar volume has the integral formula $\frac{1}{n} \int h^{-n}$; strict convexity of $t \mapsto t^{-n}$ makes it shrink under averaging. The equality cases force almost every rotated body in the mixture to be the same. Injectivity of the sine transform on even measures then forces the measure, not only its body, to be a single cross.

4 Proof

Put $K_0 = S_{\nu_E}$ and $K_R = RK_0 = S_{R\#\nu_E}$. Every $R\#\nu_E$ is even and isotropic, hence so is μ . Linearity and orthogonal equivariance of the sine transform give

$$h_{S_\mu}(x) = \int_{O(n)} h_{K_R}(x) d\pi(R). \quad (2)$$

Thus $\overline{K} := S_\mu$ is the Minkowski integral of the K_R .

For the polar inequality, the polar-coordinate volume formula and Jensen's inequality yield

$$\begin{aligned} V(\overline{K}^*) &= \frac{1}{n} \int_{S^{n-1}} \left(\int h_{K_R}(u) d\pi(R) \right)^{-n} du \\ &\leq \int_{O(n)} \frac{1}{n} \int_{S^{n-1}} h_{K_R}(u)^{-n} du d\pi(R) = V(K_0^*). \end{aligned}$$

For the primal inequality, multilinearity and continuity of mixed volume give

$$V(\overline{K}) = \int_{O(n)} V(\overline{K}[n-1], K_R) d\pi(R).$$

Minkowski's first inequality, and $V(K_R) = V(K_0)$, imply

$$V(\overline{K}[n-1], K_R) \geq V(\overline{K})^{(n-1)/n} V(K_0)^{1/n}.$$

After integration and division by the positive common factor, $V(\overline{K})^{1/n} \geq V(K_0)^{1/n}$.

It remains to check rigidity. Equality in the polar argument and strict convexity of t^{-n} imply that $h_{K_R}(u)$ is independent of R for π -almost every R and almost every u . Continuity of support functions then makes the bodies K_R identical almost surely. In the primal argument, equality makes the nonnegative gap in Minkowski's first inequality vanish almost surely. Hence almost every K_R is homothetic to $\overline{K}$. All K_R are centrally symmetric about the origin and have the same volume; under equality $\overline{K}$ has that volume as well. The homotheties therefore have ratio one and no translation, so again the K_R are identical almost surely.

Finally, the sine transform is injective on even measures [1, Proposition 2.1]. Thus equality of $K_R = S_{R\#\nu_E}$ implies that the measures $R\#\nu_E$ are identical almost surely. Equation (1) then says that μ is that cross measure. The converse is immediate. $\square$

5 A discrete criterion

Let μ be supported on finitely many antipodal lines $L_1, \dots, L_N$, and let a_j be its total mass on L_j . Form the hypergraph whose edges are the n -element subsets B of pairwise orthogonal lines (hence orthonormal bases after choosing unit representatives).

Corollary 2 (Fractional orthogonal-basis decomposition). *If there are numbers $\lambda_B \geq 0$ such that*

$$\sum_B \lambda_B = 1, \quad a_j = \sum_{B \ni L_j} \lambda_B \quad (1 \leq j \leq N),$$

then both inequalities and both equality conclusions of Theorem 1 hold for μ .

Proof. The displayed equations say exactly that μ is the convex combination, with coefficients λ_B , of the cross measures carried by the basis edges B . Apply Theorem 1. $\square$

6 Scope, upgrade attempts, and novelty

The theorem does not settle the full conjecture. A regular-simplex tight frame is isotropic but its support contains no orthogonal basis, so it cannot belong to the basis-mixture class. Moreover, that class is closed: it is the continuous image of the compact space of probability measures on $O(n)$. Thus a density argument cannot remove the obstruction.

The attack also tested the most immediate counterexample route. In dimension three, deterministic spherical sampling put the polar volume of the regular tetrahedral frame below that of the cross, and random searches over Parseval frames with four through nine lines found no violation. A separate zonoid determinant computation put the tetrahedral primal volume above the cross value. These computations are evidence only; the proof above does not use them.

A bounded literature search through 11 August 2026 found no full resolution. The complex analogue still described both the real and complex problems as open in 2015 [2]. Li–Xi–Zhang’s Grassmannian theorem proves sharp reverse inequalities when the Grassmann rank divides the dimension and explicitly leaves the non-divisible case open [3, Theorem 1.2]. The sine transform has rank $n - 1$, which does not divide n for $n \geq 3$. Exact-title, exact-conjecture, author/citation, and “basis/mixture of cross measures” searches found no prior version of the partial theorem. Novelty is therefore plausible but provisional.

References

- [1] G. Maresch and F. E. Schuster, *The sine transform of isotropic measures*, Int. Math. Res. Not. IMRN 2012 (2012), 717–739; arXiv:1207.7266.
- [2] Q. Huang and B. He, *Volume inequalities for complex isotropic measures*, Geom. Dedicata 177 (2015), 401–428.
- [3] A.-J. Li, D. Xi, and G. Zhang, *Volume inequalities of convex bodies from cosine transforms on Grassmann manifolds*, Adv. Math. 304 (2017), 494–538.